

PERICARDIAL AND RESTRICTIVE MYOCARDIAL DISEASES

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Pericardial diseases are an interesting and diverse group of disorders, and their effects on cardiac physiology are complex and fascinating. A list of the causes of pericardial disease includes nearly the entire table of contents of an internal medicine textbook because most major disease processes have the potential to affect the pericardium (Box 10.1). Despite this wide array of inciting processes, pericardial disease presents as one or more of four distinct syndromes: acute pericarditis, effusion and tamponade, pericardial constriction, and effusive-constrictive pericarditis. Syndromes may overlap, for example, when a large effusion complicates acute pericarditis or when acute inflammation tapers into a chronic process causing pericardial scarring and constriction. It is often observed that any given etiology tends to cause a predominant syndrome more frequently than others do; for instance, viral infections more commonly cause acute pericarditis, whereas malignancy usually results in effusion and tamponade. While this pattern is generally true, it should be noted that any of the conditions listed in Box 10.1 is capable of producing any of these syndromes.

This chapter focuses on the pericardial syndromes associated with hemodynamic consequences, namely, pericardial effusions, pericardial constriction, and effusive-constrictive pericarditis. In addition, because the clinical presentation and hemodynamic abnormalities of restrictive myocardial disorders behave similarly to those of constrictive pericarditis, restrictive cardiomyopathy will also be discussed, compared, and contrasted with constrictive pericarditis.

NORMAL PERICARDIAL ANATOMY AND RELATED PHYSIOLOGY

The normal pericardium consists of an inner, single layer of serous membrane known as the *visceral pericardium* and a fibrous, outer structure called the *parietal pericardium*, normally about 1 mm thick. The serous pericardium attaches to the surface of the heart and the inner surface of the parietal pericardium, defining the limits of the pericardial space; this space normally contains <50 mL of serous fluid.

Although a human can live without a pericardium without ill effects, the normal pericardium does serve specific functions.¹ The major function relates to the relatively rigid and noncompliant nature of the parietal pericardium, limiting cardiac chamber distension with changes in volume and contributing to ventricular stiffness in diastole. These properties facilitate interaction and coupling of the cardiac chambers via the interventricular septum, an important role in cardiac physiology both in health and in disease. The noncompliant nature of the pericardium results in a rapid rise in pericardial pressure with an acute accumulation of fluid in the pericardial space. The volume that can be accommodated without increasing pressure in the pericardial space is known as the *pericardial reserve volume* and is normally only about 50 to 75 mL. Slow and gradual accumulation of fluid stretches the pericardium, allowing a greater pericardial reserve volume. For this reason, relatively large volumes of fluid can collect over time without increasing pressure. Once the pericardial reserve volume is reached, the relatively steep pressure-volume curve for the pericardium mandates a rapid pressure increase in the pericardial space with any additional increments of fluid (Fig. 10.1). This explains the *last-drop* effect, in which a small increase in fluid causes a patient to deteriorate dramatically but also explains why removal of just a small amount of fluid leads to rapid improvement.²

Complex interactions exist between the pericardium, the pericardial space, the cardiac chambers, and the thoracic cavity during cardiac and respiratory cycles. The spectrum of hemodynamic abnormalities observed in pericardial diseases can best be interpreted by first understanding these important physiologic relationships.

The relationships between the cardiac chambers, the great vessels of the heart, the thoracic cavity, and the pericardial space are summarized in Fig. 10.2. Importantly, the pericardial space sits within the thoracic cavity, reflecting intrathoracic pressure (roughly -6 to 0 mm Hg). While the cardiac chambers lie within the confines of the pericardial space, the pulmonary venous system does not. The pulmonary veins exist outside of the pericardial space but within the thoracic cavity, thereby being directly influenced by changes in intrathoracic pressure. Both the inferior and superior vena cava exist outside the pericardial space and only partially within the thoracic cavity. Therefore the systemic venous circulation is mostly independent of changes in intrathoracic pressures, although a decrease in intrathoracic pressure during inspiration will augment venous filling to the right side of the heart.

In the presence of a normal pericardium, a variety of changes in cardiac physiology occur with inspiration and expiration (Fig. 10.3). During inspiration, intrathoracic pressure decreases, resulting in decreased pressure in the cardiac chambers. With little or no corresponding reduction in systemic venous pressure with inspiration, the pressure drop in the right-heart chambers augments right-heart filling. The pulmonary venous bed lies within the thoracic cavity; therefore the inspiratory reduction in intrathoracic pressure also decreases pressure within the pulmonary veins. With pressure falling uniformly in the left atrium, the left ventricle, and the pulmonary veins, no major change occurs in left-ventricular filling with inspiration. Opposite changes occur with expiration, with the net effect being a decrease in

Box 10.1 Causes of Pericardial Disease**Infectious**

Bacterial

Streptococcus, Staphylococcus, gram (–) rods

Mycobacterial

Tuberculosis

Viral

Coxsackie, echovirus, adenovirus, severe acute respiratory syndrome coronavirus 2, Epstein-Barr virus

Human immunodeficiency virus

Fungal

Aspergillosis, *Candida*, histoplasmosis

Protozoal

Amebic

Neoplastic

Primary mesothelioma

Metastatic

Breast, lung, skin, lymphoma, leukemia

Metabolic

Uremia, myxedema, amyloidosis

Immune/Inflammatory

Connective tissue disorders (lupus, scleroderma, polyarteritis nodosa, rheumatoid arthritis)

Sarcoidosis

Post-myocardial infarction, postpericardiotomy syndrome

Iatrogenic

Radiation induced

Cardiac perforation from interventional procedures

Drugs (warfarin, minoxidil, procainamide)

Trauma

Direct trauma

Aortic dissection

Congenital

Pericardial cyst

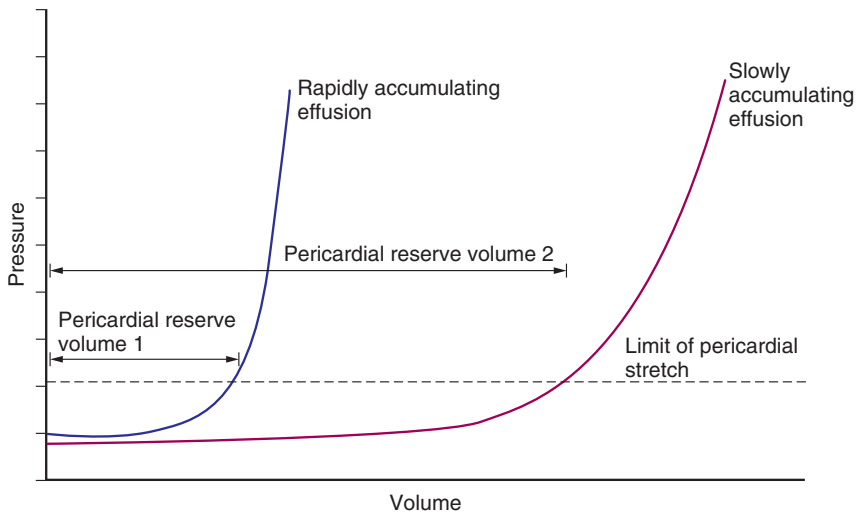
Idiopathic

Fig. 10.1 Schematic representation of the pressure-volume relationship for the pericardium. The pericardial reserve volume represents the amount of fluid that the pericardial space can accommodate before a rise occurs in the pericardial pressure. A rapidly accumulating effusion reaches the limit of pericardial stretch at a lower volume than a slowly accumulating effusion. Chronic and slow fluid accumulation allows the pericardium to stretch, creating a larger pericardial reserve volume. (Adapted from Spodick DH. Acute tamponade. *N Engl J Med*. 2003;349:684–690.)

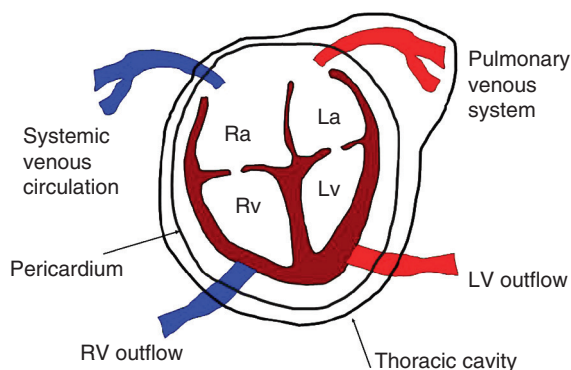


Fig. 10.2 Relationship between the cardiac chambers, the pericardial space, and the thoracic cavity. The cardiac chambers lie within the pericardial space as well as the thoracic cavity, while the pulmonary venous system sits within the thoracic cavity but outside of the pericardial space. *La*, Left atrium; *Lv*, left ventricle; *LV*, left ventricular; *Ra*, right atrium; *Rv*, right ventricle; *RV*, right ventricular.

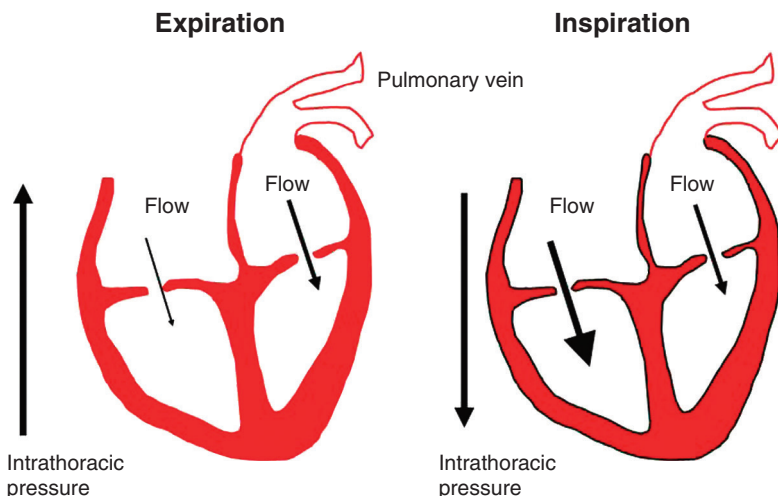


Fig. 10.3 Effects of the respiratory cycle on normal cardiac chamber filling.

right-sided filling. The net effect of inspiratory augmentation of right-ventricular filling with no change in left-ventricular filling causes a bowing of the septum from the right to the left, impairing left-sided output (Fig. 10.4). This results in a small inspiratory drop in systolic pressure known as a *pulsus paradoxus*, normally measuring <12 mm Hg and often appreciable on an arterial or aortic pressure waveform (Fig. 10.5A). The normal *pulsus paradoxus* is enhanced in patients with obesity (Fig. 10.5B).

In the presence of a normal pericardium and vacant pericardial space, the diastolic pressures of the right and left ventricles vary during the respiratory cycle independently of each other (Fig. 10.6). In contrast the systolic pressures of the right and left ventricles change during the respiratory cycle in a parallel fashion. In the presence of a normal, compliant pericardium, augmented filling of the right ventricle with inspiration is easily accommodated by the compliant right ventricle and does not lead to an increase in pressure in the right ventricle with inspiration. Instead, the changing forces with respiration cause the pressure in both chambers to decrease with inspiration and increase with expiration. Thus the right-ventricular and left-ventricular systolic pressures normally parallel each other during the respiratory cycle (Fig. 10.7). This “concordance” of right-ventricular and left-ventricular systolic pressures during the respiratory cycle is important when we consider the effects of pericardial constriction later in this chapter under the section “Hemodynamics of Pericardial Constriction.”

PERICARDIAL EFFUSIONS AND TAMPONADE

Pericardial effusions occur commonly in clinical practice from any of the multiple causes listed in Box 10.1. Although easily diagnosed with an echocardiogram, the nonspecific presenting symptoms are often attributed to other conditions

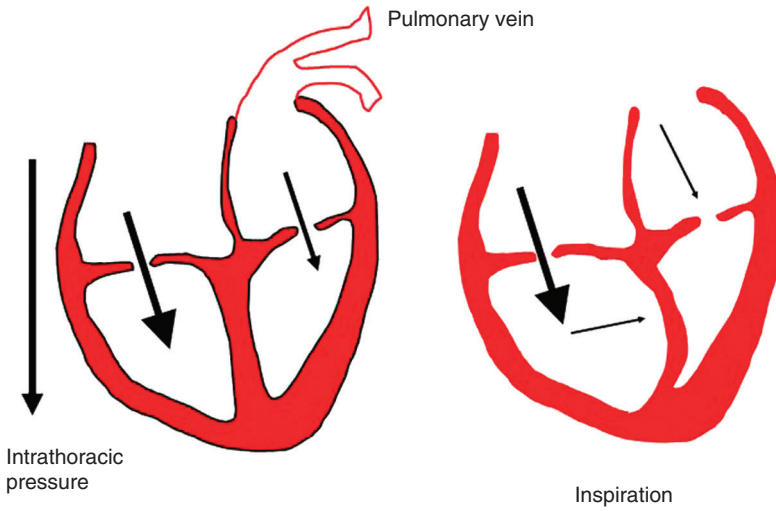


Fig. 10.4 Proposed mechanism of a pulsus paradoxus. Inspiration lowers intrathoracic pressure, resulting in no change in left-ventricular filling as both pulmonary venous pressure and left-ventricular pressure drop in parallel. However, right-ventricular filling increases with inspiration, resulting in bowing of the septum to the left (*small arrow*), impairing left-sided output.

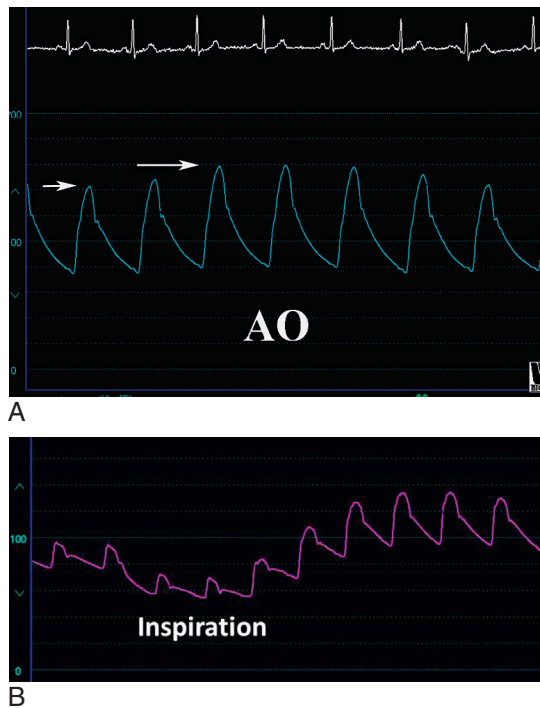


Fig. 10.5 Example of the normally observed or *physiologic* pulsus paradoxus on an aortic pressure waveform. (A) The inspiratory drop in systolic pressure, or *pulsus paradoxus*, is shown here as the difference in systolic pressure between the lowest pressure (*left arrow*) and the highest pressure (*right arrow*) and normally does not exceed 12 mm Hg. (B) A pulsus paradoxus may also occur in patients with marked obesity. AO, Aortic.

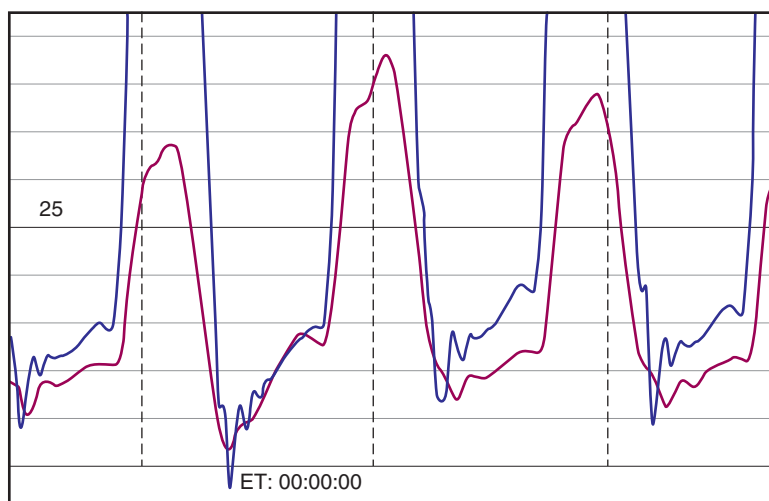


Fig. 10.6 The diastolic pressures of the right and left ventricles normally separate during the respiratory cycle.

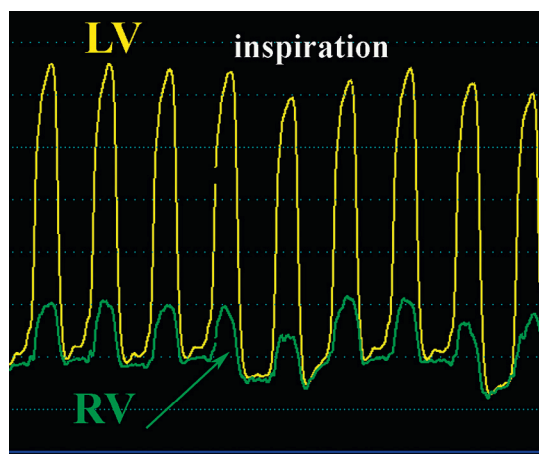


Fig. 10.7 Normally, concordance of the left-ventricular and right-ventricular systolic pressures is present during inspiration. LV, Left ventricular; RV (arrow), right ventricular.

and fail to trigger early clinical suspicion and recognition. Very frequently, this lack of suspicion during the early phases of an effusion prevents appropriate therapy, and the unrecognized effusion frequently progresses to tamponade and cardiovascular collapse, at which point the diagnosis becomes obvious. Unfortunately, the naive physician is now faced with the management of a crisis instead of basking in the joy of a clever diagnosis auspiciously treated.

Dyspnea, fatigue, and chest pain constitute the most common symptoms of an effusion. Manifestations associated with more profound hemodynamic sequelae include syncope, confusion or mental status changes, renal failure, and shock. Patients with subacute or chronic effusions may exhibit peripheral edema because of increased right-atrial (RA) pressure; similarly, patients may present with abdominal pain from hepatic distension. Physical examination nearly always reveals distended and tense neck veins. Tachycardia, arterial hypotension, and a pulsus paradoxus >12 mm Hg provide further evidence of a hemodynamically significant pericardial effusion. Two other physical examination findings are noteworthy. The Beck triad describes the constellation of elevated neck veins, hypotension, and a quiet precordium, named after the American surgeon Claude Schaffer Beck, who first described these findings in tamponade from acute, traumatic effusions. The Ewart sign describes the finding of dullness to percussion with bronchial breath sounds and evidence of lung consolidation below the left scapula seen with large, chronic pericardial effusions.³ An echocardiogram

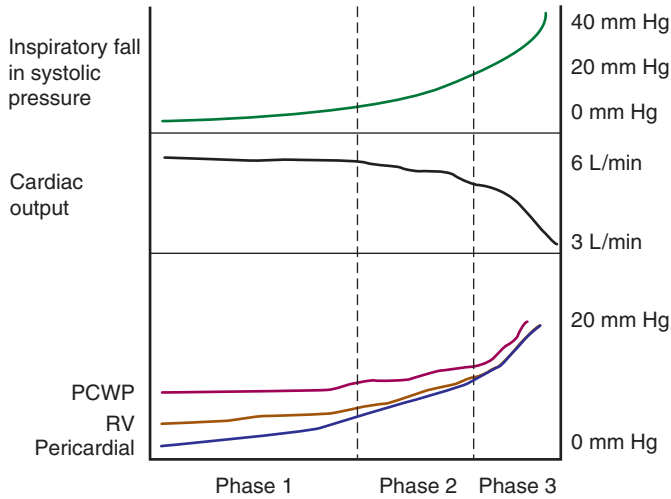


Fig. 10.8 Depiction of the various hemodynamic phases of tamponade, demonstrating the concept of the spectrum of tamponade and the effects of progressive accumulation of pericardial fluid. In phase 1, the pericardial pressure is the first to rise and leads to the elevation of the right-ventricular diastolic pressure. The pulmonary capillary wedge pressure, the cardiac output, and the inspiratory fall in systolic pressure are not yet affected. In phase 2, equalization of the diastolic pressures occurs, and cardiac output begins to fall with a greater fall in systolic pressure with inspiration evident. Phase 3 tamponade represents *classic* tamponade with a dramatic drop in cardiac output and marked pulsus paradoxus. PCWP, Pulmonary capillary wedge pressure; RV, right ventricular. (From Reddy PS, Curtiss EI, Uretsky BF. Spectrum of hemodynamic changes in cardiac tamponade. *Am J Cardiol.* 1990;66:1487–1491.)

easily identifies pericardial effusions, even in the presence of poor acoustic windows. Pericardial effusions are sometimes diagnosed surreptitiously by chest x-ray (appearing as an enlarged cardiac silhouette), chest computerized tomography (CT), or magnetic resonance imaging (MRI). The electrocardiogram is nonspecific; it may show low voltage, electrical alternans, and diffuse ST elevation consistent with pericarditis.

Once an effusion is identified, it is important to determine its hemodynamic effect. The hemodynamic sequelae of an effusion depend on the rate and volume of fluid accumulation, compliance of the cardiac chambers and the pericardium, and the filling pressures in the heart. The commonly asked question “Is there tamponade?” (usually following an echocardiographic diagnosis of an effusion) is best addressed by describing its hemodynamic impact instead of deciding whether a patient is “in tamponade.” This concept has been elegantly elucidated by Reddy and Curtiss,^{4,5} who emphasize that tamponade is not an “all or none” phenomenon, but rather represents a spectrum of hemodynamic abnormalities, beginning with isolated elevation of pericardial pressure and ending with the profound abnormalities classically attributed to tamponade.

Fig. 10.8 depicts the progressive hemodynamic derangements that occur with incremental accumulation of pericardial fluid. Initially just the pericardial pressure rises. Elevated pericardial pressure triggers compensatory mechanisms (venoconstriction and fluid retention), raising systemic venous pressure to adequately fill the right heart. This causes both RA and right-ventricular diastolic pressures to increase. During this early phase of tamponade, cardiac output is maintained and a normal inspiratory fall in systolic pressure of <10 to 12 mm Hg occurs. With additional accumulation of fluid, there is further elevation of the pericardial pressure, dragging up the pressures in the cardiac chambers during diastole and causing equilibration of pericardial, RA, and right-ventricular diastolic pressures. Left-ventricular diastolic pressure then increases, ultimately equilibrating with the right-sided diastolic pressures. This results in *equalization* of diastolic pressures across the cardiac chambers. Additional fluid accumulation drops stroke volume; cardiac output falls despite a compensatory tachycardia. The inspiratory fall in systolic pressure (pulsus paradoxus) becomes more prominent but may remain below the upper limits of normal. In the final phase of tamponade the classically recognized changes are present and include elevated and equalized diastolic pressures, a prominent inspiratory fall in systolic pressure (or pulsus paradoxus >12 mm Hg), and a precipitous drop in cardiac output and blood pressure.

HEMODYNAMICS OF PERICARDIAL EFFUSION AND TAMPONADE

Tamponade causes continuous compression of the heart throughout the cardiac cycle, preventing rapid atrial emptying when the tricuspid valve opens. This correlates with the echocardiographic finding of right-ventricular diastolic collapse and, on hemodynamic waveforms, with the absence of the y descent on the RA tracing. This abnormality occurs relatively early in tamponade (Fig. 10.9A); equilibration between RA pressure and pericardial pressure is also seen early



Fig. 10.9 (A) Loss of the y descent (arrows) usually on the right-atrial waveform in tamponade. (B) As tamponade becomes more significant, the right-atrial pressure waveform usually appears as an undulating line with no discernible a or v waves or x and y descents, and equilibration with the pericardial pressure is present. RA, Right atrial.

in this process (Fig. 10.9B). In advanced stages of tamponade the RA pressure waveform appears as an undulating and flat line without discernible a and v waves or x and y descents. Typically patients with tamponade present with elevated and equalized right-sided diastolic pressures that measure about 20 mm Hg. The reason for this is that hypotension and shock typically occur when the diastolic pressure reaches about 20 mm Hg. The patient is critically ill at this point triggering the evaluation and subsequent care (Fig. 10.10). Right-ventricular and pulmonary artery pressure waveforms often appear abnormally thin and asthenic because of reduced right-sided output from compression. Marked elevation of the pulmonary artery systolic pressure is not seen unless it is a preexisting condition. Typically pulmonary artery systolic pressure is not more than 50 mm Hg. In severe tamponade pulmonary artery systolic pressure may be only slightly higher than diastolic pressure. Pulmonary edema is generally not a feature of tamponade for poorly understood reasons but likely due to the aforementioned observation that patients present with hypotension and shock when diastolic pressures reach about 20 mm Hg, before achieving a left-atrial pressure associated with pulmonary edema. Accordingly, arterial hypoxemia should not be attributed to tamponade physiology, and its presence should prompt a search for other etiologies.

As described earlier, an inspiratory drop in systolic pressure (pulsus paradoxus) of up to 10 to 12 mm Hg is part of normal cardiac physiology. Advanced phases of tamponade exaggerate this finding, and the inspiratory fall in systolic pressure exceeds 12 mm Hg. The presence of pericardial fluid within the inelastic confines of the pericardium allows only a certain volume to fill the cardiac chambers and prevents one cardiac chamber from accommodating any additional volume without a corresponding impairment in filling of the adjacent chamber.⁶ Therefore the augmentation in right-heart filling with inspiration competes with filling of the left heart, reducing stroke volume and systolic pressure to a greater degree than seen normally.

Kussmaul first described this phenomenon over a century ago⁵:

The pulse of all arteries—with the heart movement continuing steadily—becomes very small in certain intervals that regularly occur with each inspiration, or it disappears completely, only to immediately return with expiration. I suggest to term this pulse the paradoxus pulse, in part, because of the obvious disparity between the heart action and arterial pulse, and, in part, because the pulse—though seemingly irregular—is in fact a pulse stopping or decreasing with regular repetition.

Not specific for tamponade, other conditions may exhibit a prominent pulsus paradoxus (Fig. 10.5B and Box 10.2).^{6,7} Circumstances that prevent a pulsus paradoxus despite a large and significant effusion include a coexisting atrial septal defect (because the inspiratory increase in venous return is shared between the atria), aortic regurgitation (because filling of the left ventricle is independent of respiration), or marked elevation of the left-ventricular end-diastolic pressure.

Clearly an accurate measurement of a pulsus paradoxus helps define the hemodynamic significance of an effusion. On physical examination, measurement of a pulsus paradoxus involves inflation of the blood pressure cuff above systolic pressure followed by careful auscultation during very slow deflation until any Korotkoff sound is heard with any cardiac cycle. This marks the upper limit of systolic pressure. With continued slow deflation of the cuff, the pressure at which Korotkoff sounds are heard with each cardiac cycle is noted and defines the lower limit of systolic pressure. The difference between these two recordings is the pulsus paradoxus. This procedure should be carried out while the patient is breathing normally; the patient should not be asked to “take a deep breath” while the physician frantically

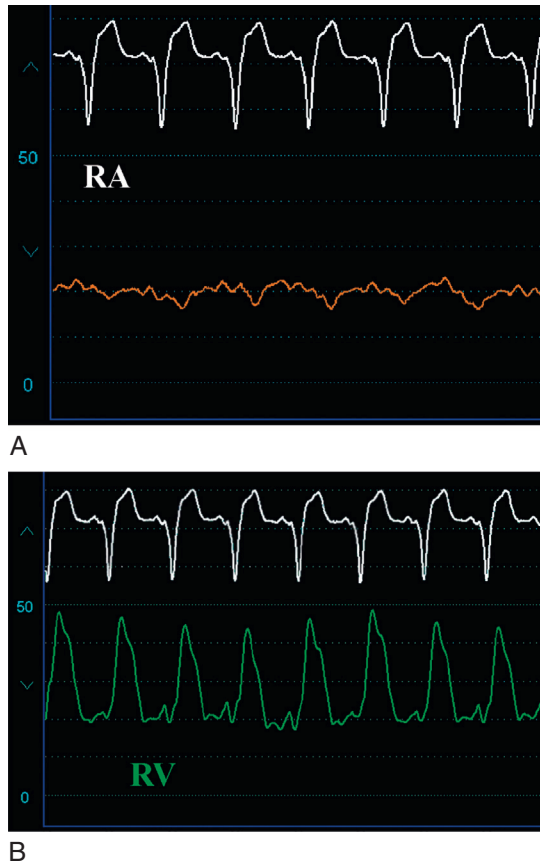


Fig. 10.10 This set of tracings was obtained from a patient with tamponade and demonstrates many of the characteristic findings. The right-atrial pressure is (A) elevated with loss of y descent, (B) equal to the right-ventricular diastolic pressure, (C) close to the pulmonary artery diastolic pressure, and (D) equal to the pulmonary capillary wedge pressure. PA, Pulmonary artery; PCWP, pulmonary capillary wedge pressure; RA, right atrial; RV, right ventricular.

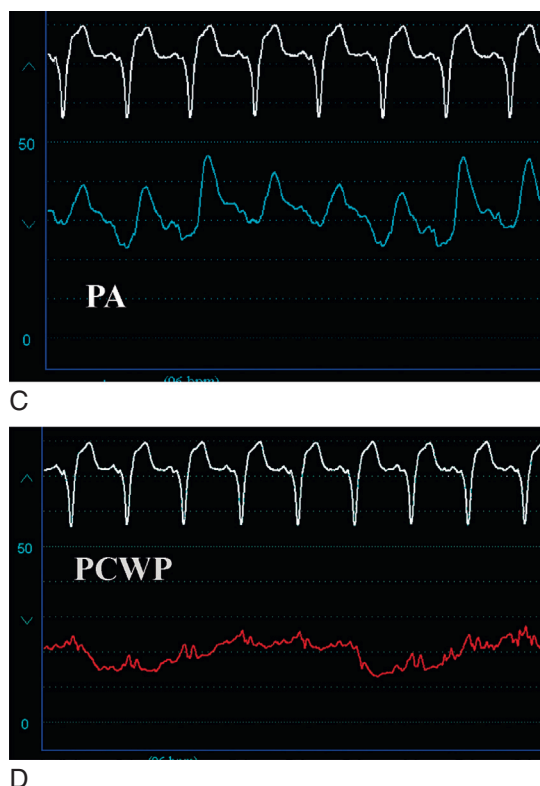


Fig. 10.10, cont'd

Box 10.2 Causes of an Inspiratory Drop in Systolic Pressure >12 mm Hg (Pulsus Paradoxus)

- Pericardial tamponade
- Effusive-constrictive pericarditis
- Right-ventricular infarction
- Asthma
- Chronic obstructive pulmonary disease
- Congestive heart failure
- Obesity
- Ascites
- Pregnancy
- Pulmonary embolism
- Tension pneumothorax

inflates a blood pressure cuff. Arterial pressure recordings identify a pulsus paradoxus with greater sensitivity than physical examination (see Fig. 10.5). In tamponade the pulsus paradoxus may be dramatic, completely obliterating the pulse, as described by Kussmaul (Fig. 10.11).

The classic hemodynamic findings of tamponade may be obscured or absent in several conditions. Right-ventricular hypertrophy and pulmonary hypertension prevent an effusion from fully compressing the right ventricle during diastole. Accordingly, early diastolic filling may not be impaired; the y descent on the RA waveform may appear normal despite tamponade (Fig. 10.12). In the setting of severe pulmonary hypertension, the effects of the effusion may only be apparent on the RA waveform after pericardiocentesis is performed and the fluid removed (Fig. 10.13). Patients with preexisting elevation of left-ventricular diastolic pressure may not have equalization of left-ventricular diastolic or pulmonary capillary wedge pressure (PCWP) with right-sided chamber diastolic pressures. Diastolic pressures may not



Fig. 10.11 The arterial pressure waveform in this tracing obtained from a patient with tamponade demonstrates total pulsus paradoxus with complete obliteration of the arterial pressure with inspiration. AO, Aortic; d, diastolic pressure; s, systolic pressure.

equalize across all chambers if there is marked pulmonary hypertension despite advanced tamponade, or if there is a loculated effusion that selectively compresses just the right heart. This latter scenario is most common in a postoperative effusion.⁸ As noted earlier, a pulsus paradoxus may be absent in the presence of an atrial septal defect, significant aortic regurgitation, elevated left-ventricular end-diastolic pressure, or localized tamponade.

Elevated and equalized right-sided diastolic pressures characterize classic tamponade. However, a pericardial effusion may cause serious hemodynamic compromise, despite low (6–12 mm Hg) diastolic pressures. This so-called *low-pressure tamponade* was initially described predominantly in the setting of hypovolemia.^{2,9} It may be seen with acute tamponade, particularly from hemorrhage into the pericardium, as may occur from iatrogenic perforation during a coronary intervention or electrophysiologic procedure (Fig. 10.14). A diminished intravascular volume from hemorrhage into the pericardium prevents the aforementioned compensatory increase in venous pressure further exacerbating the consequences of cardiac compression, manifest by cardiovascular collapse, despite *low* right-sided pressures.

The syndrome of low-pressure tamponade has not been properly defined or systematically studied. A study that involved 143 patients with an effusion who underwent careful hemodynamic evaluation at the time of pericardiocentesis provides important information regarding this syndrome.¹⁰ The authors' proposed criteria first define the presence of tamponade as equalization of intrapericardial and RA pressure with right-sided transmural pressure <2 mm Hg. Low-pressure tamponade is then defined as the presence of an intrapericardial pressure <7 mm Hg before pericardiocentesis and RA pressure <4 mm Hg after intrapericardial pressure has been lowered to near 0 mm Hg by pericardiocentesis. Using this definition, low-pressure tamponade was present in 20% of patients who underwent pericardiocentesis. These patients were less likely to exhibit the classic clinical signs of tamponade; the presence of pulsus paradoxus was present in only 7% of patients. The frequency of symptoms, effusion size by echocardiography, and preexisting use of diuretics were not different from patients with classic tamponade.¹⁰

TREATMENT OF TAMPONADE

Small to moderate-sized effusions without significant hemodynamic sequelae often do not require any specific treatment. It is important to prevent dehydration and avoid vasodilator drugs in these patients because of their dependence on preload. In the presence of significant hemodynamic consequences or symptoms, the effusion needs to be removed promptly. Hypotension and cardiac output can be improved by a rapid infusion of normal saline (500 mL over 10 minutes)¹¹ and by the use of intravenous pressor agents while preparing for pericardiocentesis. Removal of the



Fig. 10.12 Patients with pulmonary hypertension and right-ventricular hypertrophy often have preservation of the *y* descent on the right-atrial pressure waveform, despite tamponade, as shown in this example of a patient with tamponade from postpericardiotomy syndrome after mitral valve replacement for chronic mitral regurgitation. The patient had known pulmonary hypertension prior to surgery, with pulmonary pressure in the 70-mm Hg systolic range. (A) Pressures obtained prior to pericardiocentesis.

pericardial fluid should proceed immediately in clinically unstable patients. The treatment of choice in most patients is percutaneous pericardiocentesis.^{1,2} Compromised patients with loculated effusions or in whom pericardiocentesis was not successful need surgical drainage, often accomplished by the creation of a pericardial window. Finally percutaneous balloon pericardiotomy has been used to reduce the chance of recurrent effusions in patients with malignant pericardial effusion.^{12,13}

Pericardiocentesis is often performed in the catheterization laboratory but can be accomplished at the bedside in a critically ill patient who is rapidly deteriorating from the effusion or cannot be transported to the catheterization laboratory because of clinical instability. Echo guidance identifies the ideal site for percutaneous access to the effusion and determines whether loculation is present. A right-heart catheterization procedure performed in conjunction with pericardiocentesis allows an assessment of hemodynamics before and after fluid removal. The goal of pericardiocentesis is to remove all fluid and achieve a pericardial pressure of zero (or less). Inability to obtain a pericardial pressure of zero indicates loculation (Fig. 10.15). After successful pericardiocentesis, the right-heart pressure measurements should show a reduction in right-sided diastolic pressures and a return of the *y* descent on RA pressure waveforms. Fluid analysis may provide insight regarding the etiology of the effusion, particularly in the case of bacterial infection or malignancy.

Echocardiography is an important tool in the diagnosis and management of pericardial effusion. In addition to identifying the presence of an effusion, echocardiography can also illuminate the hemodynamic consequences of the effusion and is helpful in guiding therapeutic pericardiocentesis. The echocardiographic findings in patients with tamponade are beyond the scope of this textbook and have been well described elsewhere.¹⁴

PERICARDIAL CONSTRICTION

Constrictive pericarditis can be caused by any of the disorders that affect the pericardium (shown in Box 10.1). In undeveloped nations in Asia, Africa, and the Middle East, tuberculosis remains a common cause. In Europe and



Fig. 10.12, cont'd. (B) Pressures obtained after successful pericardiocentesis show that the average pericardial pressure was reduced to <0 mm Hg; the right-atrial pressure waveform remains abnormal with a more prominent y descent. (C) Pressures also obtained postpericardiocentesis; compare this to (A) and note the greater prominence of the y descent. RA, Right atrial; RV, right ventricular.

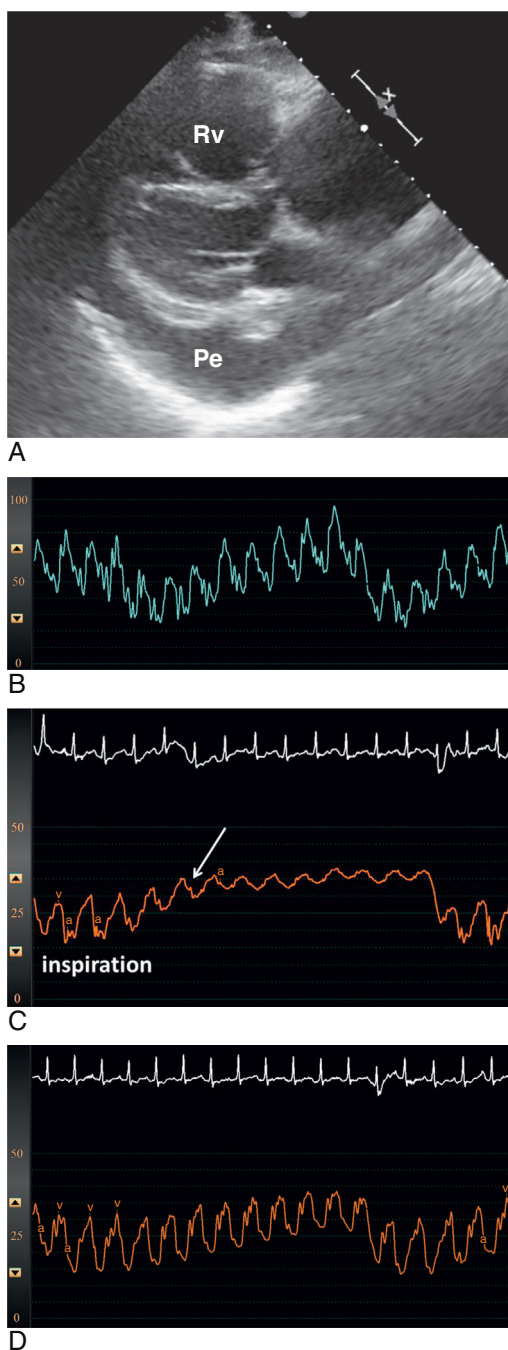


Fig. 10.13 These tracings were obtained from a 55-year-old female with pulmonary hypertension caused by chronic thromboembolic disease who subsequently developed a large pericardial effusion. (A) An echocardiogram revealed marked enlargement of the right ventricle and right atrium with a large pericardial effusion. (B) There is marked respiratory variation and severe elevation of the pulmonary artery pressure (80/50 mm Hg end expiration). (C) The right-atrial pressure is markedly elevated with a mean pressure at end expiration of approximately 35 mm Hg. Note that there is preservation of the y descent (*arrow*) and this becomes more prominent with inspiration. Tamponade usually obliterates the descents with no respiratory variation. (D) After pericardiocentesis, the y descents on the right-atrial pressure become very prominent and consistent with underlying severe tricuspid regurgitation. *Pe*, Pericardial effusion; *Rv*, right ventricle.

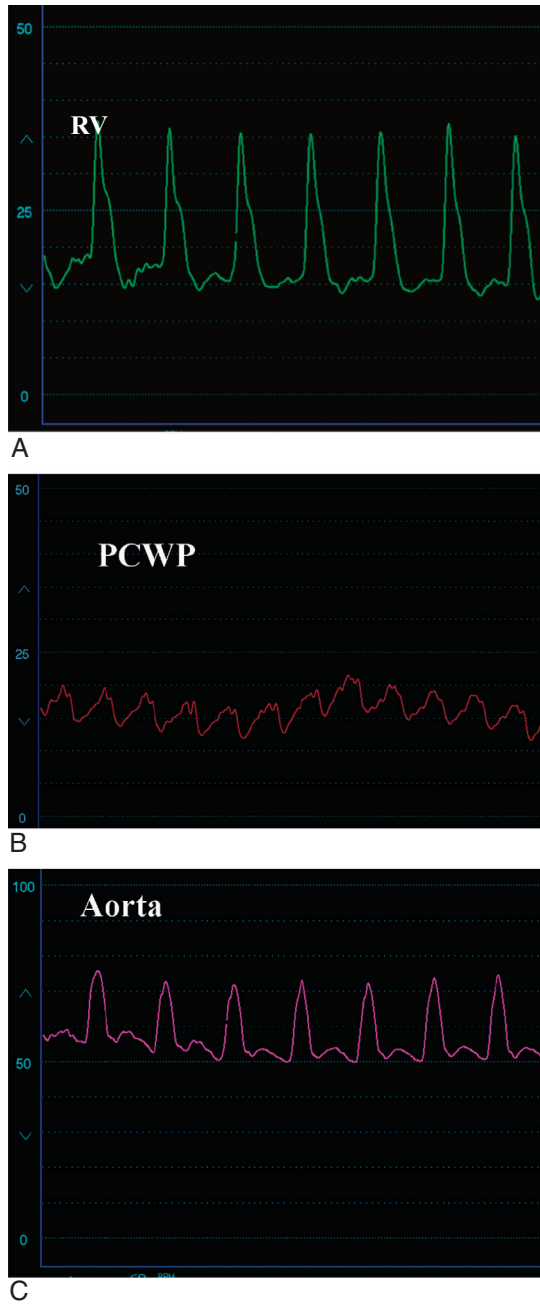


Fig. 10.14 Low-pressure tamponade in a patient with acute tamponade from coronary perforation during percutaneous coronary intervention. The patient developed acute hypotension during an attempt at opening a chronically occluded left anterior descending artery. (A) The right-ventricular diastolic pressure was only mildly elevated at 13 mm Hg, and (B) the pulmonary capillary wedge pressure was only 16 mm Hg; (C) however, profound hemodynamic collapse occurred with marked hypotension and tachycardia. *PCWP*, Pulmonary capillary wedge pressure; *RV*, right ventricular.

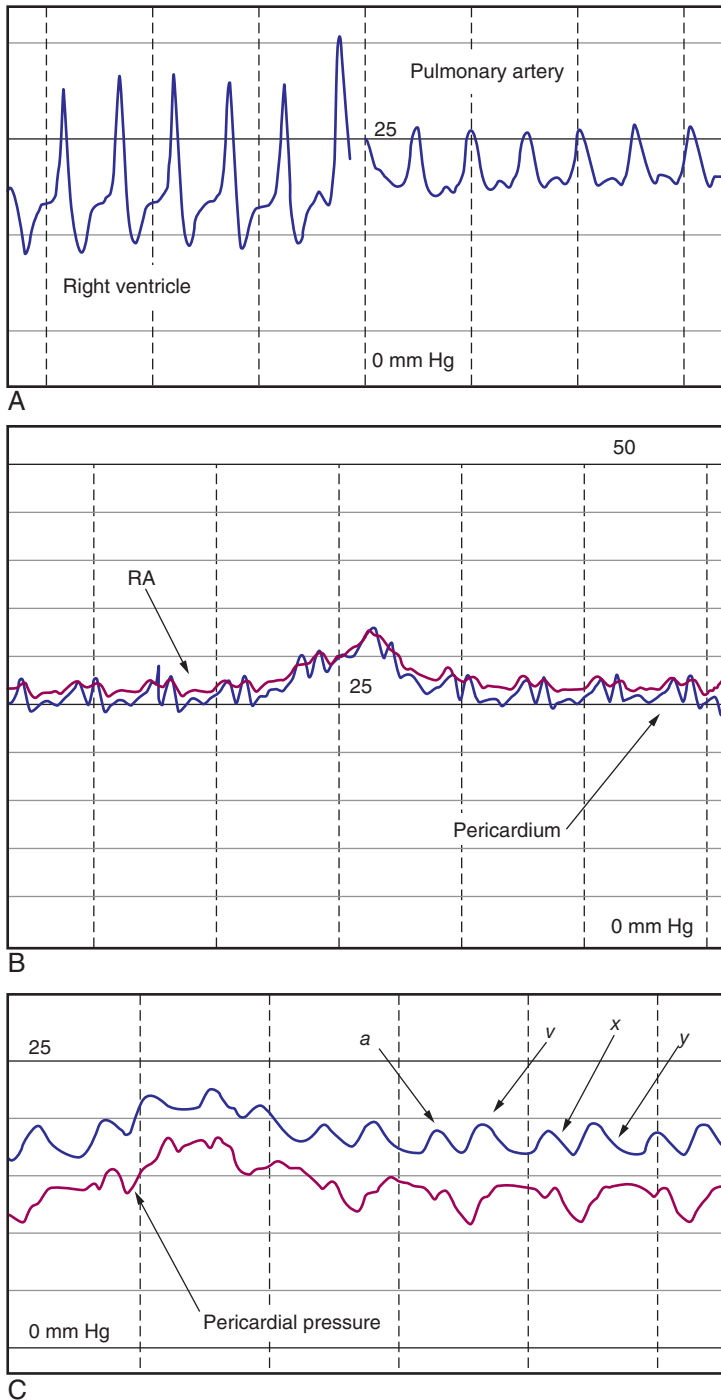


Fig. 10.15 Loculation of fluid is suggested by the inability to reduce the pericardial pressure to zero or less with pericardiocentesis. In this example, (A) typical tamponade is present with equalization and elevation of the right-ventricular end-diastolic and pulmonary artery pressure and (B) the right-atrial and pericardial pressures. (C) Following pericardiocentesis, elevation of the pericardial pressure is still present. Note that evidence exists for some hemodynamic improvement with a return of the x and y descents postpericardiocentesis. *a*, a wave; *RA*, right atrial; *v*, v wave.

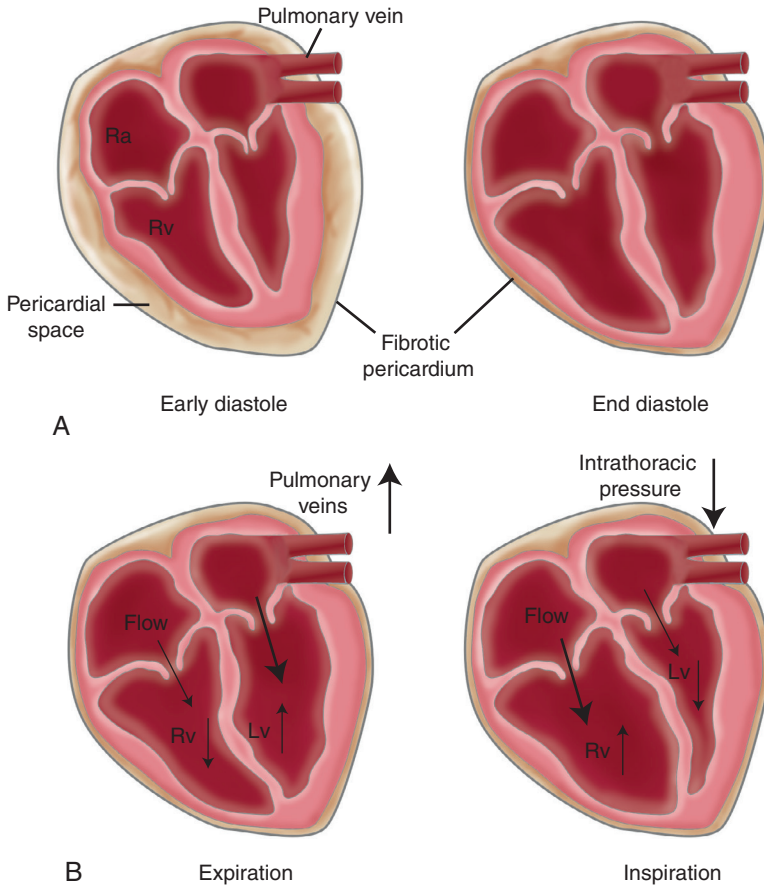


Fig. 10.16 Schematic representation of the effects of pericardial constriction on cardiac chamber physiology. The rigid pericardium encases the heart and defines the volume that can be accommodated by the cardiac chambers. Early diastolic filling is relatively unimpaired; (A) filling abruptly ceases by mid-diastole when the volume determined by the pericardial space has been reached. (B) The effect of pressure changes during the respiratory cycle in the presence of pericardial constriction. *Lv*, Left ventricle; *Ra*, right atrium; *Rv*, right ventricle.

North America, tuberculosis is rare, accounting for <5% of cases with the great majority of cases (>80%) due to idiopathic causes, prior chest irradiation (usually for Hodgkin lymphoma or breast cancer), postoperative from prior heart surgery, connective tissue disorders, and infectious causes of pericarditis. Viral pericarditis only rarely (<1% of cases) progresses to constriction but bacterial pericarditis resulting in purulent pericarditis may commonly (20%–30% of cases) progress to constriction.^{15–18}

In contrast to effusions and tamponade, constrictive pericarditis presents as a chronic illness. With relatively nonspecific and often vague symptoms of dyspnea, fatigue, cachexia, edema, and abdominal distension, constrictive pericarditis is often misdiagnosed as cirrhosis of the liver, malignancy, or other chronic disease. Physical exam signs reflect the physiologic abnormalities. Elevation of the neck veins with a prominent *y* descent is virtually a *sine qua non* of constrictive pericarditis and can also be a useful method to distinguish this entity from cirrhosis.¹⁸ Edema and ascites are present in the majority of patients. Auscultatory findings include an absent apical impulse with a pericardial knock. With inspiration, the jugular venous pressure does not decrease normally, either remaining elevated or increasing further (Kussmaul sign). Pulsus paradoxus is rare in pure forms of constriction but may be seen in cases of effusive constriction or in the setting of coexisting pulmonary disease.¹⁹

PHYSIOLOGY OF PERICARDIAL CONSTRICTION

In pericardial constriction a rigid, unyielding shell encases the heart, creating important hemodynamic consequences.^{15,20} First, the constraining pericardium sets a limit to the total volume of blood able to enter the cardiac

chambers, seriously affecting ventricular filling during diastole. In early diastole the ventricle fills briskly. By mid-diastole, however, the chamber volume defined by the rigid pericardium is reached and filling halts abruptly, rapidly raising pressure. In fact, early diastolic filling is more rapid than normal because of the chronically underfilled state of the ventricle. A second important hemodynamic consequence relates to the increase in ventricular interdependence from the constraining pericardium. Without a compliant pericardium to buffer the changes in volume and pressure that occur during the cardiac and respiratory cycles, volume and pressure changes that occur in one chamber will be reflected in the adjacent chamber. Finally, the rigid, constraining pericardium isolates the pericardial space from changes in intrathoracic pressure during respiration. Thus inspiration lowers intrathoracic pressure but not cardiac chamber pressure, increasing venous filling into the thorax but not the right-heart chambers. This causes an increase or no change in jugular venous pressure with inspiration instead of the expected decrease (Kussmaul sign). In addition, inspiration lowers pulmonary venous pressure, but the negative inspiratory pressure is not transmitted to the left ventricle, causing a decrease in the transmitral gradient with inspiration and a corresponding diminished filling of the left ventricle with a drop in left-ventricular systolic pressure. Reduced filling of the left ventricle along with increased ventricular interdependence results in a corresponding increase in right-ventricular filling and an associated increase in right-ventricular systolic pressure. The opposite effects occur with expiration. Accordingly, the normally parallel fall and rise in right-ventricular and left-ventricular systolic pressure with inspiration and expiration become discordant in constrictive pericarditis. These physiologic effects are summarized in Fig. 10.16. The finding of discordance of right-ventricular and left-ventricular systolic pressures with inspiration is highly sensitive and specific for constrictive pericarditis.^{21,22}

HEMODYNAMIC FINDINGS IN CONSTRICTIVE PERICARDITIS

The physiologic abnormalities associated with constrictive pericarditis result in several classic hemodynamic findings (Box 10.3). Several abnormalities are apparent on the RA waveform. The RA pressure shows little respiratory variation. The *v* wave is often greater than the *a* wave due to noncompliance of the atrium from the encasing pericardium. The rapid atrial emptying that occurs early in diastole along with increased atrial pressure and the relatively underfilled state of the ventricle cause a rapid *y* descent on the RA waveform (Fig. 10.17). These are often described as an *M* or *W* pattern. The ventricular waveforms exhibit a characteristic *square root sign* or “dip and plateau” during diastole from the abrupt cessation in filling from the constraining pericardium (Fig. 10.18). The left-ventricular and right-ventricular diastolic pressures do not separate by more than 5 mm Hg. Pulmonary hypertension is rare; the pulmonary artery and right-ventricular systolic pressures rarely exceed 50 mm Hg. However, none of these hemodynamic abnormalities are specific for constrictive pericarditis. Several other conditions can cause similar findings, termed *constrictive physiology* (Box 10.4). Restrictive cardiomyopathy shares many of these features, and tricuspid regurgitation can be indistinguishable from constriction by these hemodynamic criteria.²³ Examples of *pseudoconstriction* are shown in Figs. 10.19 and 10.20.

The most valuable hemodynamic findings for the diagnosis of constrictive pericarditis relate to the dynamic respiratory effects described earlier. Dissociation of intrathoracic and intracardiac pressures can be observed in the cardiac catheterization laboratory by demonstrating respiratory variation in the pressure gradient between the PCWP (that reflects the pulmonary vein) and the left ventricle in early diastole. In constriction, because the pulmonary veins sit outside the confines of the pericardial space and are subject to changes in intrathoracic pressure, a fall occurs in the early diastolic gradient with inspiration and a subsequent rise occurs with expiration (Fig. 10.21). More importantly, however, is the finding of discordance in left-ventricular and right-ventricular systolic pressures with inspiration, a sign of increased ventricular interdependence (Figs. 10.22 and 10.23). In one study, discordance in left-ventricular and right-ventricular systolic pressures was the most sensitive (100%) and specific (95%) sign for constrictive pericarditis.²¹

Importantly, these dynamic respiratory effects on ventricular pressure require meticulous attention to detail, high-quality hemodynamic tracings, and a regular rhythm.^{22,24} Ideally these subtle effects are best assessed with a respirometer to record inspiration and expiration, and high-fidelity micromanometer catheters rather than the fluid-filled catheters in clinical use, especially for the demonstration of the left-ventricular and PCWP gradient. Recognizing that this equipment is not available in most catheterization laboratories, the demonstration of left-ventricular to

Box 10.3 Summary of Hemodynamic Findings Observed in Constrictive Pericarditis

- Elevated right-sided diastolic pressures
- M* or *W* pattern on right-atrial waveform with rapid *y* descent
- v* greater than *a* wave due to noncompliance of the right atrium
- Little to no respiratory variation in right-atrial pressure
- Square root sign or dip and plateau pattern on ventricular pressure waves
- Failure to separate left-ventricular and right-ventricular diastolic pressures >5 mm Hg
- Ratio of right-ventricular end-diastolic pressure to right-ventricular systolic pressure >1:3
- Right-ventricular (or pulmonary artery) systolic pressure <50 mm Hg
- Pulmonary capillary wedge pressure declines greater than left-ventricular early diastolic pressure during inspiration
- Discordance in left-ventricular and right-ventricular systolic pressure with inspiration



Fig. 10.17 The right-atrial pressure is elevated, and the y descent is rapid and prominent in pericardial constriction, as shown in a tracing from a patient with idiopathic constrictive pericarditis proven at surgery. RA, Right atrial.

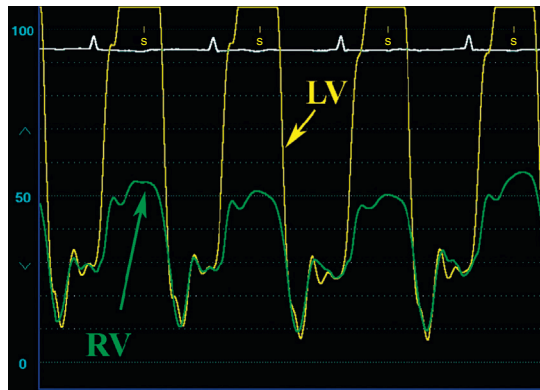


Fig. 10.18 The right-ventricular and left-ventricular pressure waveforms characteristically demonstrate a diastolic dip and plateau or square root sign due to unimpeded early diastolic filling followed by a rapid rise in pressure by mid-diastole from pericardial constraint. The right-ventricular and left-ventricular diastolic pressures usually do not separate by more than 5 mm Hg in pericardial constriction. These findings are shown in this tracing from a patient with idiopathic constrictive pericarditis. LV, Left ventricular; RV, right ventricular.

Box 10.4 Causes of “Constrictive” Hemodynamics (Pseudoconstriction)

- Restrictive cardiomyopathy
- Right-ventricular infarction
- Pulmonary hypertension
- Obesity
- Severe tricuspid regurgitation
- Acute volume overload
- Acute heart failure
- Acute severe mitral regurgitation

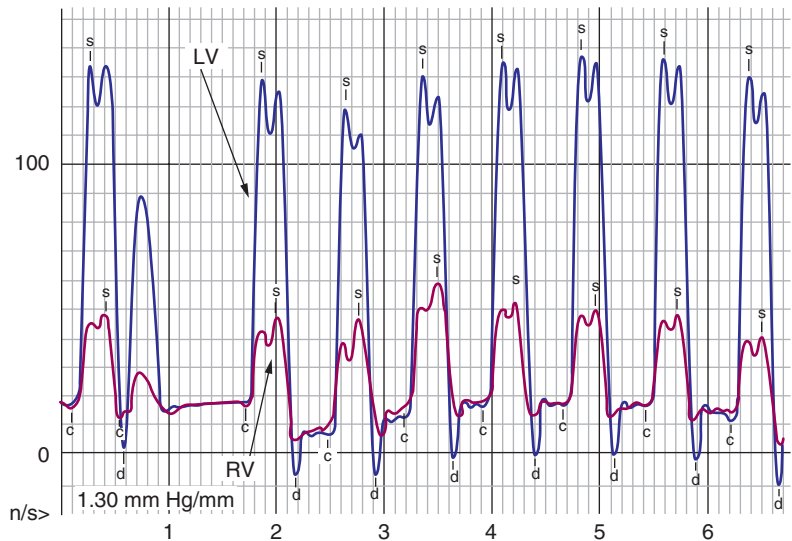


Fig. 10.19 Some of the hemodynamic findings in constrictive pericarditis may be seen in other disorders. For example, the tracing shown was obtained from a 47-year-old male with severe end-stage lung disease, who was under evaluation for lung transplantation and had a normal pericardium on a computerized tomography scan. Pulmonary hypertension is present with a pulmonary artery systolic pressure of 50 mm Hg. The right-ventricular and left-ventricular diastolic pressures demonstrate a square root sign and diastolic pressures within 5 mm Hg. However, concordance exists between the right-ventricular and left-ventricular systolic pressures. *c*, End-diastolic pressure; *d*, early diastolic pressure; *LV*, left ventricular; *RV*, right ventricular; *s*, systolic pressure.

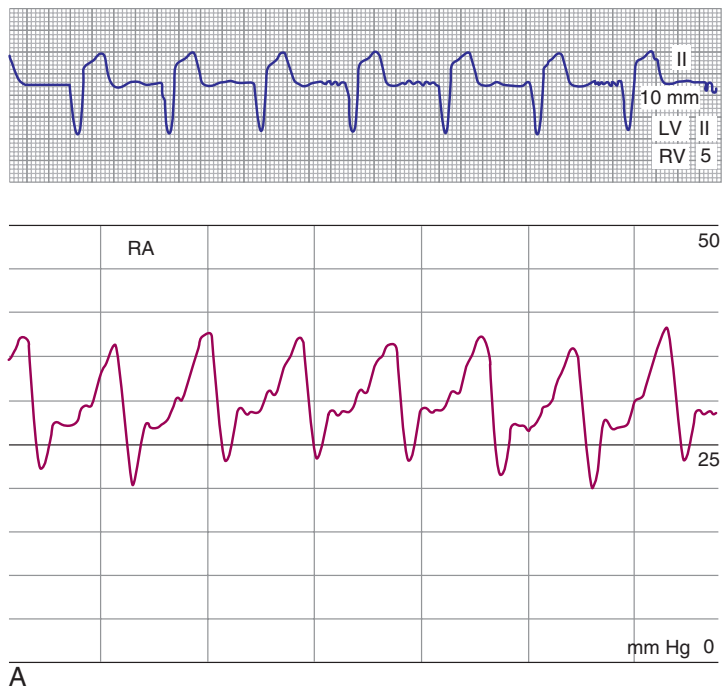


Fig. 10.20 (A) Severe tricuspid regurgitation may lead to hemodynamic findings similar to pericardial constriction with a square root sign, elevated and equal right-ventricular and left-ventricular diastolic pressures and (B) prominent *y* descent on the right-atrial waveform. The prominent *y* descent, however, is due predominantly to a large *v* wave. *LV*, Left ventricular; *RA*, right atrial; *RV*, right ventricular.

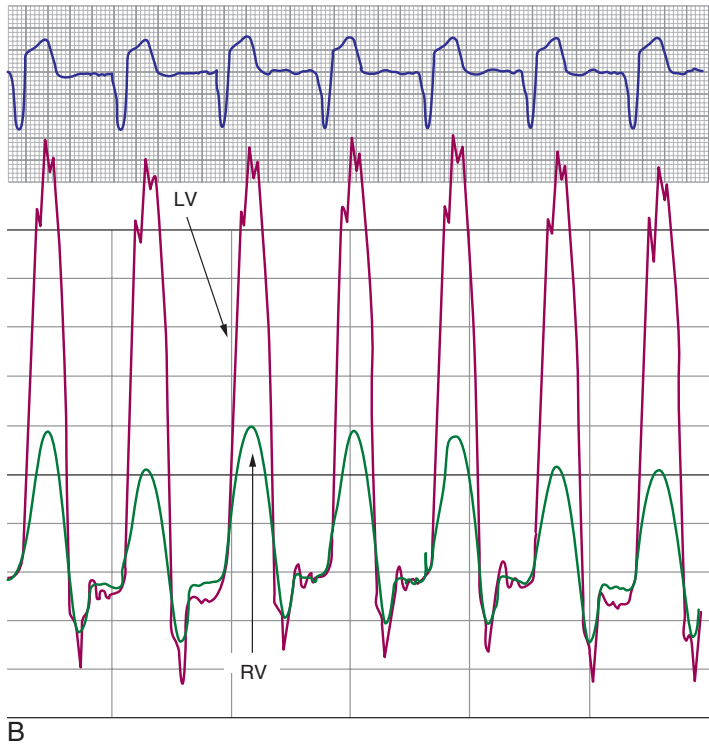


Fig. 10.20, cont'd.

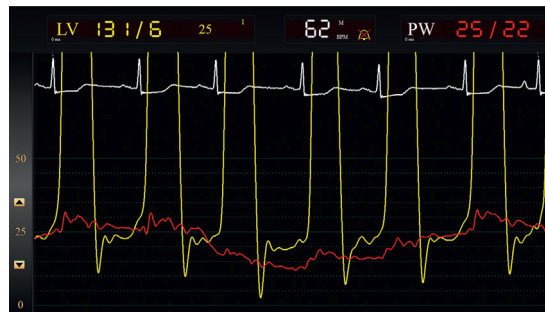


Fig. 10.21 Demonstration of dissociation of intrathoracic and intracardiac pressures in a patient with pericardial constriction. There is a decrease in the pulmonary capillary wedge pressure during inspiration that is not paralleled by the left-ventricular pressure, which remains essentially unchanged. There is a subsequent rise in pulmonary capillary wedge pressure with expiration.

right-ventricular systolic discordance remains a valuable clue. Be aware, however, that the rhythm should be regular to detect these changes. Patients with atrial fibrillation or other causes of irregular rhythm should have their rhythm regularized with a temporary pacemaker during the hemodynamic assessment to avoid the confusing effect on pressure with varying RR intervals. Another manifestation of ventricular interdependence in constriction is shown in Fig. 10.24. In constrictive pericarditis, the area under the right-ventricular systolic pressure curve increases during inspiration while the area under the left-ventricular systolic pressure curve decreases (Fig. 10.24)²². In restrictive cardiomyopathy or other causes of heart failure, during the inspiratory phase, the area under the right ventricular and left-ventricular systolic pressure curves decrease in a concordant fashion.

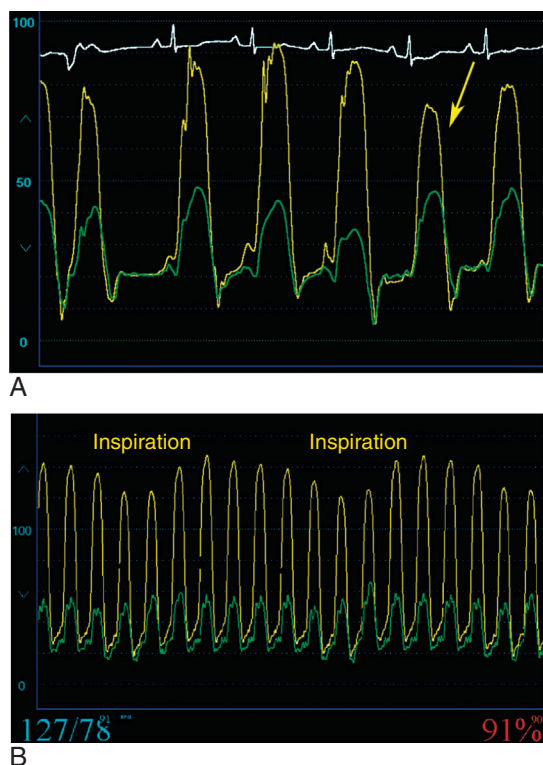


Fig. 10.22 Examples of ventricular interdependence in pericardial constriction. (A) A drop in left-ventricular systolic pressure with inspiration and a rise in right-ventricular systolic pressure (arrow). (B) Panel A is shown over several respiratory cycles.

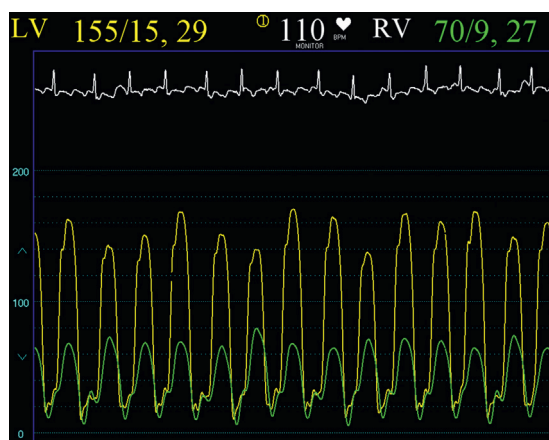


Fig. 10.23 This patient had constrictive pericarditis and demonstrates discordance of the right-ventricular and left-ventricular systolic pressures during the respiratory cycle. This patient has underlying atrial flutter, but the rhythm is regular. *LV*, Left ventricular; *RV*, right ventricular.

Ultimately, a final diagnosis of constrictive pericarditis is a clinical one based on the incorporation of appropriate signs and symptoms coupled with demonstration of constrictive physiology by catheterization, echocardiography, or other imaging modalities, including CT scan and MRI.^{14–16,18–20,25} The astute clinician should be aware that the syndrome of constrictive pericarditis may still exist in the absence of global pericardial involvement, and focal or asymmetric

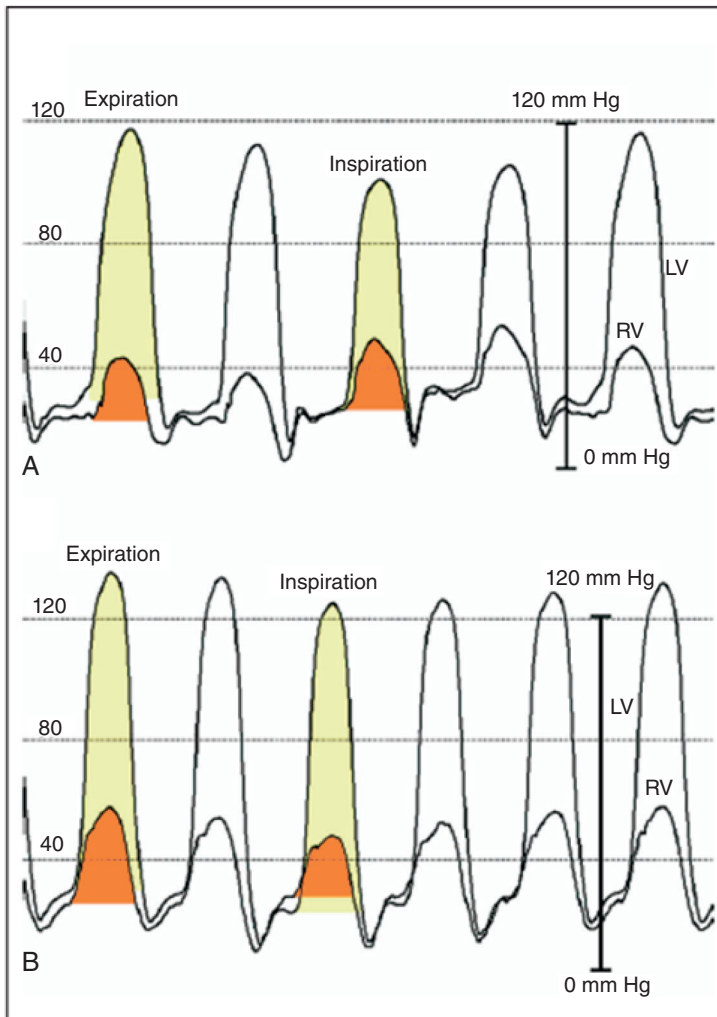


Fig. 10.24 Simultaneous left (LV) and right (RV) ventricular pressure tracings obtained with a high-fidelity manometer during inspiration and expiration. The tracings shown in the top panel (A) are from a patient with constrictive pericarditis. Inspiration results in an increase in the area of the RV pressure curve (orange) and a decrease in the area of the LV pressure curve (yellow). The opposite occurs during expiration. The tracings shown in the bottom panel (B) are from a patient with restrictive cardiomyopathy. During inspiration, there is a decrease in the area under the RV pressure curve (orange) compared to expiration, while the area under the LV pressure curve (yellow) remains unchanged during the respiratory cycle. (From Talreja DR, Nishimura RA, Oh JK, Holmes DR. Constrictive pericarditis in the modern era: novel criteria for diagnosis in the cardiac catheterization laboratory. *J Am Coll Cardiol*. 2008;51(3):315–319).

thickening may cause significant hemodynamic compromise.²⁶ Finally, pericardial calcification on chest x-ray or fluoroscopy is present in only a minority of cases of constriction (Fig. 10.25).

Constrictive pericarditis can be treated medically with diuretics if only mild symptoms and relatively low RA pressure are present. Chronic elevations of the RA pressure may lead to hepatic cirrhosis. Therefore if medical therapy is chosen, it is important to assess the response to therapy by reassessing RA pressure. Many patients with constriction will require surgical pericardiectomy, a procedure that is effective at lowering RA pressure and improving symptoms.^{16,17}

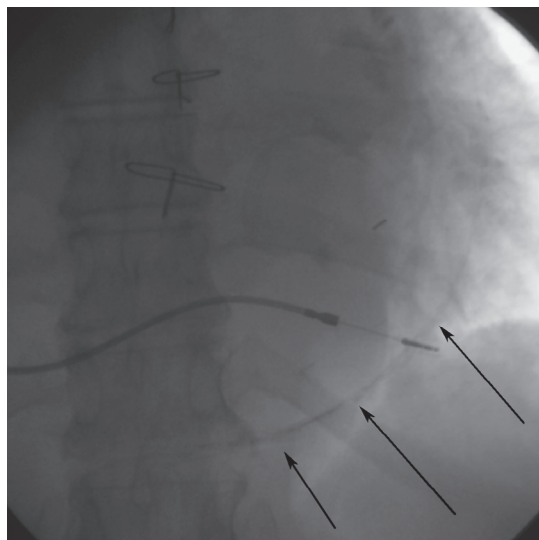


Fig. 10.25 Pericardial calcification seen on fluoroscopy (arrows) in a patient with pericardial constriction. Although this is a helpful diagnostic tool, pericardial calcification is only rarely observed.

EFFUSIVE-CONSTRICTIVE PERICARDITIS

Some patients with pericarditis, initially found to have compressive physiology in the presence of the effusion, continue to have marked hemodynamic abnormalities when the effusion is removed. In pure tamponade the removal of fluid by pericardiocentesis achieves a pericardial pressure of zero or less with a return to normal, right-sided hemodynamics. Persistent elevation in RA pressure, despite removal of all fluid and achievement of a pericardial pressure <0 mm Hg, suggests continued abnormality of the pericardium, usually caused by inflammation and a constricting effect of the visceral pericardium. In addition to the elevation in pressures, the RA waveform typically demonstrates a prominent and rapid y descent similar to constriction. This rare entity, known as *effusive-constrictive pericarditis*, is present in approximately 8% of patients with tamponade who undergo catheterization.²⁷ Numerous etiologies have been associated with this syndrome, including postoperative pericarditis, neoplastic disease, idiopathic, and infectious pericarditis.^{15,27} In many of these patients the constrictive component resolves as the inflammatory process subsides, but this syndrome may ultimately lead to chronic constriction and the need for pericardiectomy.²⁷ The typical hemodynamic finding that suggests effusive-constrictive pericarditis is shown in Fig. 10.26. In this case acute tamponade complicated postpericardiotomy syndrome several weeks after mitral valve replacement surgery. Initially, hemodynamics suggested classic tamponade with pericardial pressure equal to RA pressure and blunting of the y descent on the RA waveform. After removing all fluid and obtaining a pericardial pressure of zero, the RA pressure remained elevated, now with a rapid and prominent y descent consistent with constriction. Another example of effusive-constrictive pericarditis in a patient with purulent pericarditis is shown in Fig. 10.27.

RESTRICTIVE CARDIOMYOPATHY

Restrictive cardiomyopathy refers to a group of uncommon disorders of the heart muscle, resulting in impaired ventricular filling with normal systolic function. Increased ventricular stiffness causes noncompliance of the ventricles and elevated chamber pressures during diastole with only small increases in volume. The condition can affect either or both ventricles and may not uniformly affect both ventricles.

Amyloidosis, prior chest irradiation, and idiopathic restrictive cardiomyopathy are probably the most common causes outside of the tropics. The most common cause in the world is endomyocardial fibrosis, but this is confined to tropical regions. Other conditions that lead to restrictive cardiomyopathy are rare (Box 10.5).^{28,29} The clinical presentation and hemodynamic abnormalities are very similar to constrictive pericarditis.^{28–30} In fact, many of the hemodynamic findings described for constriction may be present in restrictive cardiomyopathy, particularly the square root sign and the prominent y descent on the RA waveform.

Differentiation between constrictive pericarditis and restrictive cardiomyopathy in the catheterization laboratory may be difficult (Figs. 10.28–10.30). Features more consistent with restriction include separation of the ventricular diastolic pressures >5 mm Hg, the presence of pulmonary artery pressure >50 mm Hg, and a ratio between

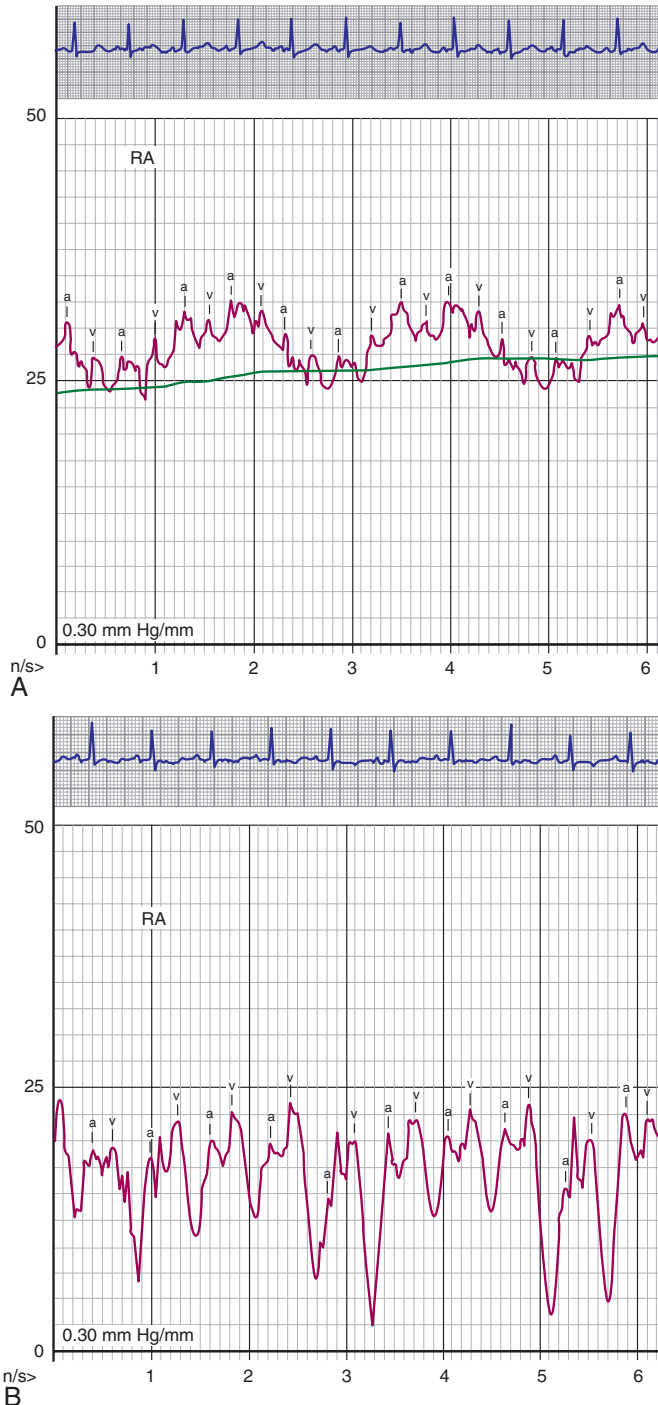


Fig. 10.26 Effusive-constrictive pericarditis in a patient who presents with tamponade from postpericardiotomy syndrome. (A) Prior to pericardiocentesis, right-atrial pressure was elevated with attenuated *x* and *y* descents. Following pericardiocentesis, the pericardial pressure was reduced to 0 mm Hg with no residual pericardial fluid discernible on echocardiography. (B) The postpericardiocentesis right-atrial pressure, however, remained elevated with a prominent *y* descent consistent with effusive-constrictive pericarditis. *a*, *a* wave; *RA*, right atrial; *v*, *v* wave.

right-ventricular diastolic pressure and right-ventricular systolic pressure of $>1:3$. As noted earlier, discordance in right-ventricular and left-ventricular systolic pressures with inspiration is the most valuable method of distinguishing the two entities in the cardiac catheterization laboratory. There are several echocardiographic features that are useful to distinguish restriction from constriction, and these have been described in detail elsewhere.^{15,16,25,31} Other clinical features more consistent with restriction include the presence of biatrial enlargement and atrioventricular valve regurgitation. The presence of pericardial calcification favors pericardial constriction but is rare. Electrocardiogram abnormalities, such as low voltage and conduction delay, are more common in restriction than constriction. MRI and CT scans are helpful in identifying the thickened pericardium of constriction. Endomyocardial biopsy might be considered if there is a strong suspicion of restriction. However, constrictive pericarditis and restrictive cardiomyopathy may coexist in patients with radiation-induced heart disease and hypereosinophilic syndrome. Ventricularization of the RA waveform

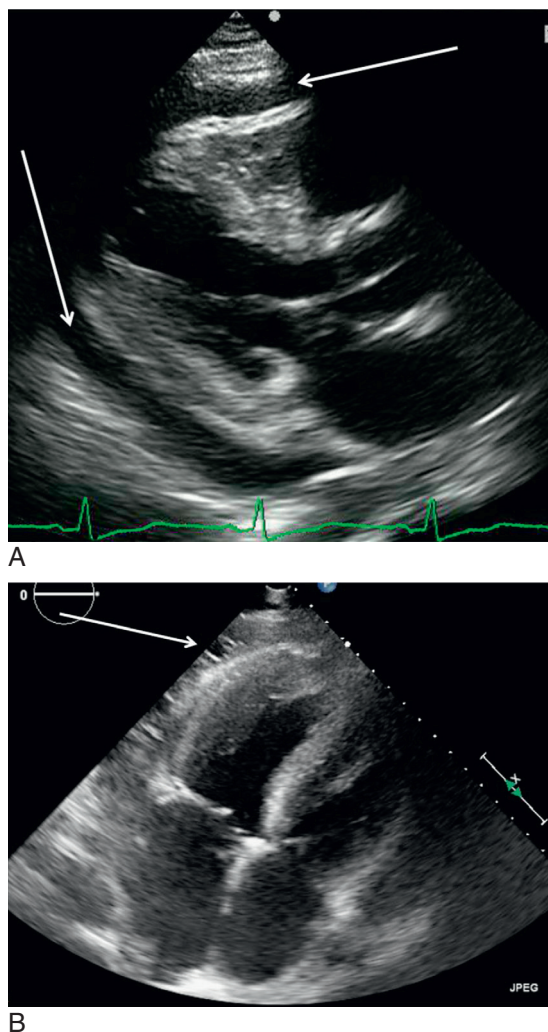


Fig. 10.27 These images are from an unfortunate 53-year-old male with end-stage renal disease who was on hemodialysis and developed progressive shortness of breath after having a febrile illness for 1 week. (A) He was found to have a pericardial effusion on echocardiography (arrows) without tamponade. Methicillin-resistant *Staphylococcus aureus* subsequently grew in his bloodstream. (B) He was then brought to the catheterization laboratory for pericardiocentesis to determine if the effusion was infected, but the effusion was noted to be highly loculated with fibrinous strands (arrow), and only a very small amount of purulent fluid was removed. This fluid subsequently grew *S. aureus*.

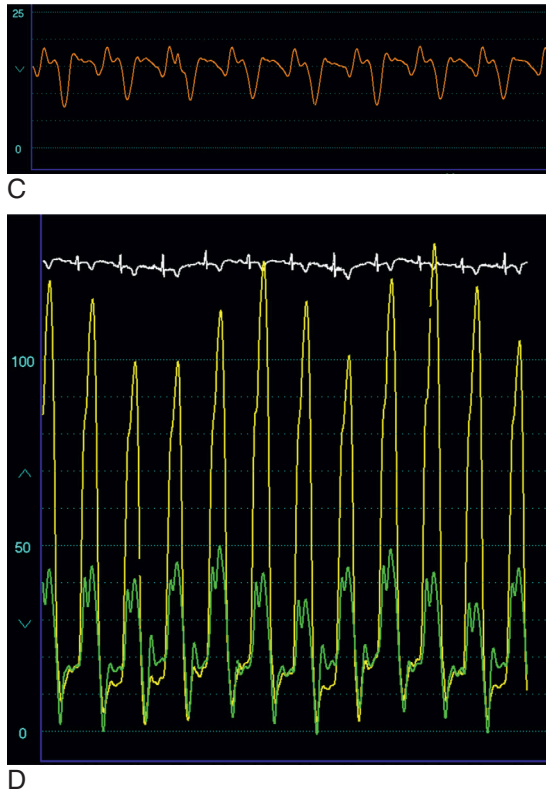


Fig. 10.27, cont'd. (C) A right-heart and left-heart catheterization showed evidence of constrictive physiology with elevated right-atrial pressure and prominent y descent and (D) evidence of discordance between right-ventricular and left-ventricular systolic pressures. The patient was treated with a surgical window to drain the purulent effusion and prolonged antibiotics and subsequently improved with resolution of the constrictive physiology.

Box 10.5 Some Causes of Restrictive Cardiomyopathy

- Cardiac amyloidosis
- Idiopathic
- Hemochromatosis
- Hypertrophic cardiomyopathy
- Sarcoidosis (rare; usually dilated cardiomyopathy)
- Gaucher disease
- Fabry disease
- Friedreich ataxia
- Glycogen storage disease
- Hypereosinophilia
- Carcinoid
- Transplant rejection
- Prior radiation therapy
- Neoplasm
- Drug induced (antimalarials)
- Endomyocardial fibroelastosis (most common in world but in tropical nations)

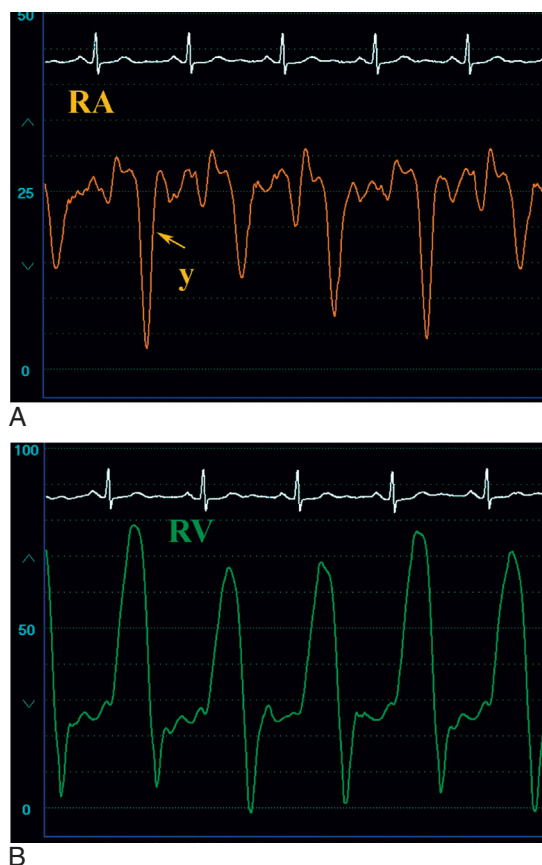


Fig. 10.28 Hemodynamics obtained from a 47-year-old patient with ascites, edema, and left-ventricular hypertrophy on echocardiography with normal systolic blood pressure, normal systolic function, and biatrial enlargement with a normal pericardium on a computerized tomography scan. (A) Note the prominent y descent on the right-atrial waveform and the (B) square root sign on the right-ventricular waveform.

in the absence of tricuspid regurgitation has been associated with amyloid restrictive cardiomyopathy.³² Finally, restrictive cardiomyopathy significantly elevates levels of brain natriuretic peptide compared to the normal levels measured in constriction.³³

HEMODYNAMIC CONSEQUENCES OF PRIOR CHEST IRRADIATION

Radiation therapy used to treat malignancies such as Hodgkin disease and breast cancer can result in cardiac complications many years after treatment in survivors. In addition to radiation-induced coronary artery disease, conduction system disease, and calcification of the intrathoracic aorta, radiation may also cause pericardial, myocardial, and valvular heart disease.³⁴

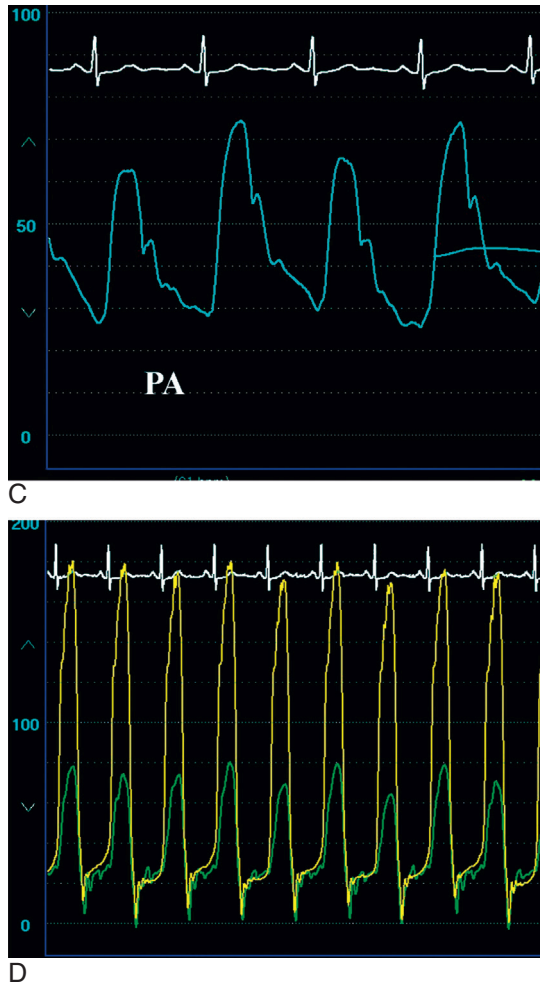


Fig. 10.28, cont'd. (C) The pulmonary artery pressure exceeds 50 mm Hg, and although the left-ventricular and right-ventricular diastolic pressures are within 5 mm Hg, there does not appear to be (D) ventricular interdependence, as the systolic pressures are concordant. *PA*, Pulmonary artery; *RA*, right atrial; *RV*, right ventricular.

Radiation can cause acute pericarditis and effusions at the time of the treatment; however, effusions may also occur years after the exposure. In addition, chronic constrictive pericarditis may develop and occur even in the absence of earlier acute pericarditis. Effects on the myocardium include the development of interstitial fibrosis, which causes diastolic heart failure and restrictive cardiomyopathy. Radiation-induced injury to the heart valves may cause regurgitation or stenosis to the aortic and/or mitral valves. It is not uncommon to see a combination of these conditions in any one patient, creating a very challenging problem for the surgeon. For example, the 50-year-old female with prior chest irradiation shown in Fig. 10.31 had involvement of both the aortic and mitral valves but also had severe calcification of the ascending aorta as a consequence of irradiation and thus was not an operative candidate.

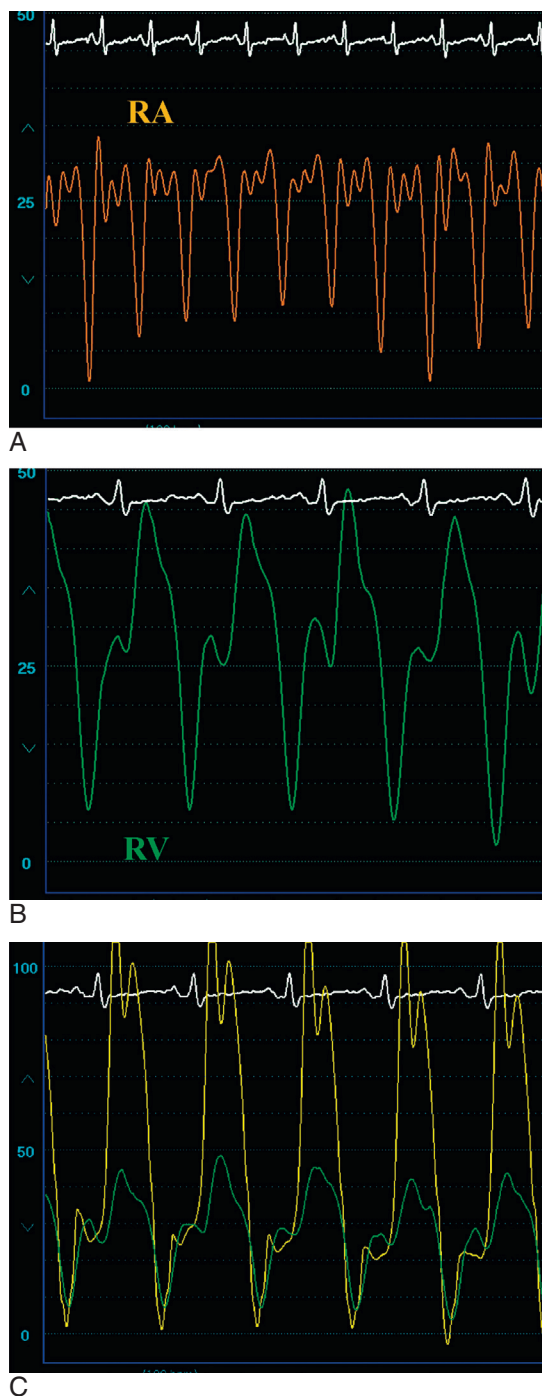


Fig. 10.29 Restrictive hemodynamics in a patient with severe rejection after heart transplantation. (A) Right-atrial pressure is markedly elevated with prominent y descent; and (B) right-ventricular pressure shows a dip and plateau pattern. (C) Some separation between left-ventricular and right-ventricular diastolic pressures was apparent, and no evidence of ventricular interdependence was present. *RA*, Right atrial; *RV*, right ventricular.

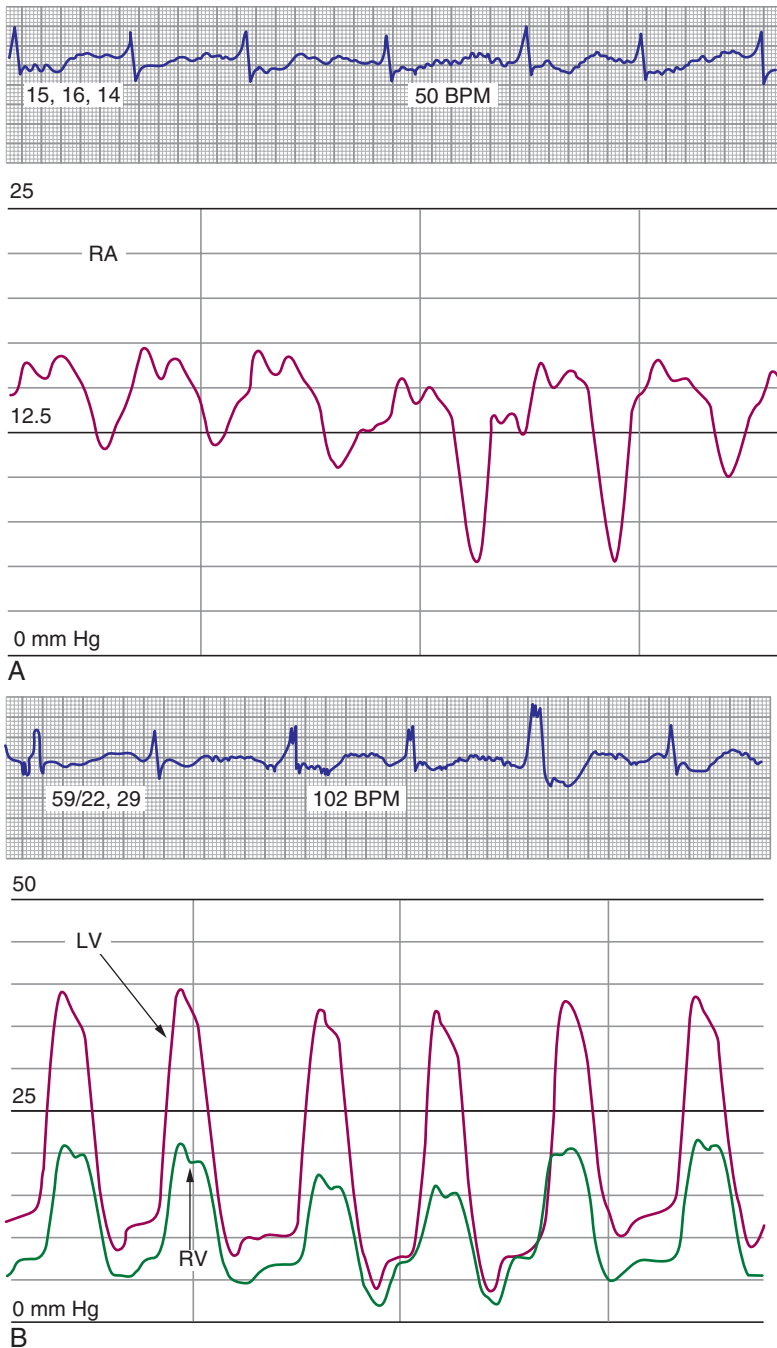


Fig. 10.30 Restrictive cardiomyopathy from biopsy-proven amyloidosis from multiple myeloma. (A) Note the prominent y descent on the right-atrial waveform. (B) Note the separation between right-ventricular and left-ventricular diastolic pressures and the concordance between systolic pressures. *BPM*, Beats per minute; *LV*, left ventricular; *RA*, right-atrial; *RV*, right ventricular.

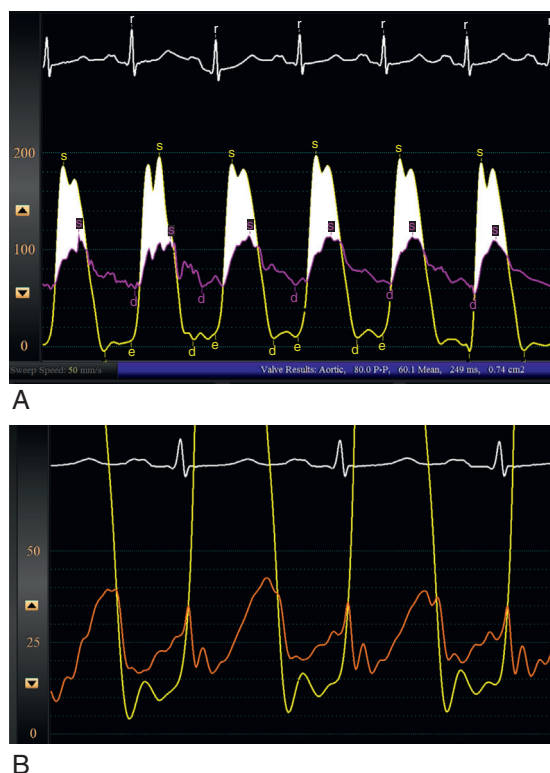


Fig. 10.31 These hemodynamic tracings were obtained from a 50-year-old female with prior chest irradiation for Hodgkin disease 30 years earlier. (A) Simultaneous left-ventricular and aortic pressure show severe aortic stenosis with a mean gradient of 60 mm Hg. (B) Simultaneous left-ventricular and left-atrial pressures show moderate mitral stenosis. *d*, Early diastolic pressure; *e*, end-diastolic pressure; *s*, systolic pressure.

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